

Puzzle Hand Analyzer - How to Use

This tool watches a speed-puzzling video and measures how the puzzler's hands move: left vs. right activity, finger work, grip/piece-handling pace, rhythm, fatigue over the session, and time spent sorting vs. assembling.

Everything runs on your own computer. Your videos are never uploaded anywhere.

FOR THE PUZZLER (easy, no typing)

One-time setup (about 5 minutes, done once)

1. Install Python (the engine this tool runs on):
2. **Windows / Mac:** download from <https://www.python.org/downloads/> and run the installer.
3. **Windows users:** on the first installer screen, tick the box "**Add Python to PATH**" before clicking Install. This matters.
4. **Linux:** Python 3 is usually pre-installed. The launcher will prompt you to install tkinter (and venv) from your package manager if they're missing - it tells you the exact command.
5. Keep all the files in this folder together. Don't separate them.

Every time you want to analyze a video

1. Open the tool:
2. **Windows:** double-click **Start_Windows.bat**
3. **Mac:** double-click **Start_Mac.command** (If Mac says it's from an unidentified developer: right-click the file, choose Open, then Open again. Only needed the first time.)
4. **Linux:** in a terminal in this folder, run `./Start_Linux.sh` (or double-click it if your file manager is configured to execute shell scripts). If the launcher reports that tkinter is missing, it prints the exact one-line install command for your distro (e.g. `sudo apt install python3-tk python3-venv` on Ubuntu/Debian). Run that, then start the launcher again.
5. The first launch installs a few things and is slow. Later launches are quick. A small window will open.
6. **Choose a video** - click "Choose video file..." and pick your puzzle video.
7. **(Optional) Mark the board** - in the preview picture, **click and drag** to draw a box around the puzzle board. Then adjust it:
 8. **Drag any corner** to resize the box
 9. **Drag the middle** of the box to move it around
10. **Drag outside** to draw a fresh box (replaces the old one)
11. Click **Clear board** to start over This unlocks "sorting vs. assembling" stats. If you're not sure, just skip it.

12. Options:

13. "Left/Right labels look reversed" - leave unticked for now. If the results come out with the hands swapped, run it again with this ticked.
14. "Annotated review video" - makes a video with the hand tracking drawn on, so you can check the analysis looks right. Default is the **first 2 minutes** (a quick sanity check, small file). Pick a longer length, **Full video**, or **Don't create** as you prefer.
15. "Faster processing" - quicker, slightly less precise.
16. Click **Analyze video**. A long video can take 10-20 minutes. You can leave the window open and come back. Progress shows in the box.
17. When it finishes, it offers to open the **results folder**. The results are saved next to your video, in a folder ending in `_analysis`.

What you get

- `..._report.pdf` / `..._report.png` - **start here**: a single-page session summary with the headline numbers, per-hand comparison, pace over time, and bimanual breakdown.
- `..._heatmap.png` - 6 mini heatmaps showing where the hands worked in each sixth of the session. A green box on each panel marks the auto-detected "active zone" - where pieces were actually picked up or placed in that window - so you can see how it tightens as the puzzle gets assembled.
- `..._timeline.png` - second-by-second chart of hand movement, finger work, and individual grip cycles across the whole session.
- `..._session.png` - per-segment pace, rhythm, and zone time.
- `..._metrics.json` - all the numbers as a single file (send to whoever is reviewing the data).
- `..._calibration.jpg` - a single frame with the hands labeled. Check this if you ever doubt Left/Right.
- `..._annotated.mp4` - the review video (length set in Options).
- `..._perframe.csv` - the raw data, openable in Excel.

If the Left and Right hands are swapped

Open `..._calibration.jpg`. If the labels are on the wrong hands, run the analysis again with "Left/Right labels look reversed" ticked. Once you know the right setting for your camera, it stays the same for every video from that setup.

FOR ADVANCED USERS (command line)

The engine is `puzzle_hands.py` and can be run directly:

```
python puzzle_hands.py VIDEO0.mp4
python puzzle_hands.py VIDEO0.mp4 --board 0.42,0.20,0.92,0.78
python puzzle_hands.py VIDEO0.mp4 --segment-seconds 60 --no-video
```

```
python puzzle_hands.py VIDEO.mp4 --fps 8          # faster
python puzzle_hands.py VIDEO.mp4 --swap-hands
```

Run `python puzzle_hands.py --help` for all options. The board rectangle is in 0-1 coordinates - read it off the grid in `..._calibration.jpg`.

TROUBLESHOOTING

- **"Python is not installed"** - install it from python.org (Windows: tick "Add Python to PATH"), then run the launcher again.
- **The launcher window flashes and closes** - open a terminal / command prompt in this folder and run the launcher from there to see the error message.
- **It's very slow** - that's normal for long videos. Tick "Faster processing", or analyze a shorter clip first.
- **Hands swapped** - see the section above.
- **Best results** come from a steady overhead camera with both hands and the table clearly visible.

REQUIREMENTS

Python 3.10-3.12 recommended. The launcher installs the rest (mediapipe, opencv, numpy, matplotlib) automatically on first run.